Surfing explained by Newtonian physics.

A novel explanation of surfers nose-riding and riding a wave.

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Fig A. Nose-riding [20] due to the Coanda effect.

Abstract

For the first time, Newtonian mechanics is applied to explain the different forces generated (Force = ma) surfing, as described below:

- **Nose-riding** and how a surfer can stand at the tip of the board. See Fig. A. The rear part of the surfboard re-directs waterflow upwards due to the Coanda effect. This action causes a change in momentum of the water flow and generates an upward turning force. The equal, and opposite downward reactive force pulls the surfboard toward the water. See Fig. A.
- **Riding a wave** and how a surfboard can travel laterally across a wave, faster than the wave's speed forward towards the beach. See Fig. B. The rising mass of water of a wave constantly pushes the surfer upwards, like an upward-moving escalator. If the surfboard is pointed downwards, then it can accelerate diagonally across the constantly rising front part of the wave. See Fig. B.



Fig B. Riding a wave. [2]

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1. INTRODUCTION

A. *Theories of surfing.*

The physics of surfing is an enigma. There is no proven explanation of surfing which is supported by adequate experimentation. Surfing has evolved by intuition, as well as trial-and-error. This approach makes it difficult to learn how to surf and to design optimal equipment.

Current discussion of surfing appears to be misguided and based on concepts from fluid mechanics (i.e. Bernoulli or hydrodynamics) and non-Newtonian physics. This paper argues that fluid mechanics can help explain fluid flow, but Newtonian mechanics (Force = ma) explains the resultant forces acting on the surfboard generated by the fluid flows.

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2. BACKGROUND – SURFING

A. Wave basics.

A transverse sea wave is a vertical oscillation of the water, representing a horizontal movement of energy through the water. The wave's upward movement is caused by water moving in a circle as the energy passes through it.

It is not accurate that only the energy moves in a wave, and that water particles do not move. The water particles do move – in a circle to create the up and down movement of the wave at the water's surface. It is more accurate to say that there is no net movement of water particles over time. See Fig. 2a-i.

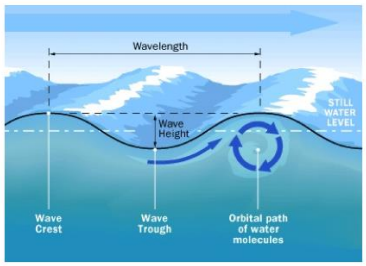


Fig. 2a-i. Waves circulate the water as they travel. [9]

Close to the beach the wave breaks then recedes, providing surfing opportunities. See Fig. 2a-ii.

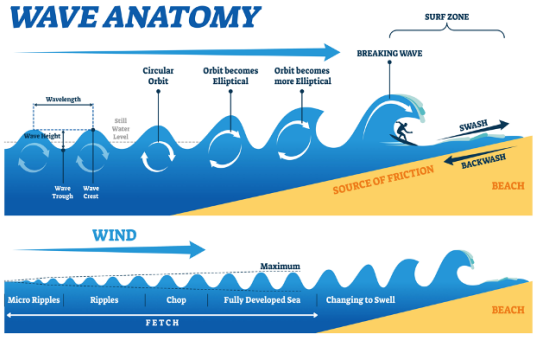


Fig. 2a-ii. Waves circulate the water. [9]

Typical features of waves include:

- Wave direction is approximately perpendicular to the beach. The waves usually hit the beach in even lines.
- Wave lengths, amplitudes and speeds vary depending on the causes of the waves and circumstances (e.g. beach geography, wind, weather, ….).
- Surface waves are most commonly caused by wind.

As waves get closer to the beach their amplitudes (height) increases, wavelength shortens, speed decreases, and their shape becomes more elliptical (as compared to circular). The increase in wave height can cause the wave to crest. This is because as the water gets shallower and the wave's energy dissipates close to the beach.

B. Surfboard basics.

Surfboards are found in various different shapes, sizes and materials. The surfboard's length, width, fins, front tip, and back end vary a lot. See Fig. 2b-i.

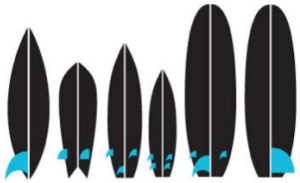


Fig. 2b-i. Some surfboard designs.

These aspects affect how water flows around the surfboard, and therefore, the forces generated. Specifically, these aspects affect the mass of water displaced each second (m/dt) and the velocity to which this water is accelerated by the board (dv); which determine the Newtonian forces generated (Force = ma = m/dt * dv). In turn, the forces generated affect the speed, maneuverability, and stability of the surfboard.

Surfboards materials also, which affects their buoyancy and interaction with the water. The size of the surfboard affects its buoyancy, how well it floats in the water.

Beginner surfers prefer large (wide and long), thick (ie. buoyant) surfboards with a concave 'A' shaped underside and big fins, as these surfboards are slow and stable (eg. Long-board). Whereas, expert surfers prefer relatively small (i.e. narrow and short), thin, and unstable surfboards with tapered fonts and tails, because these are fast and maneuverable (e.g. Short-board or Fish).

The surfboard undersides have different shapes, which can vary across the length of the surfboard. This affects the fluid flow around the surfboard (i.e. m/dt and dv), and therefore, the forces generated. See Fig. 2b-ii.

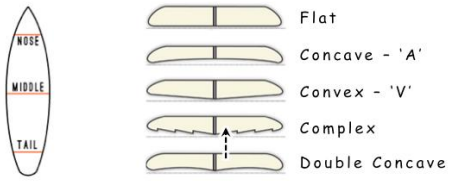


Fig. 2b-ii. Examples of different surfboard undersides.

In broad terms, a convex 'V' shape on the underside of a board adds to stability. This is because it displaces water downwards, towards the center of the surfboard. Any shift in balance or orientation of the board creates forces that push the surfboard back to its original, stable position.

A concave 'A' underside board shape increases instability. This is because it displaces water downwards, away from the center of gravity in the middle of the surfboard. It is the opposite of a 'V' shaped board. Any shift in balance or orientation of the board creates forces that exaggerate the movement away from the board's balanced position. This difference in board shape is a similar concept to the anhedral ('A') and dihedral ('V') shaped wings of an airplane.

C. The Coanda effect

Fluid flow naturally follows a curved surface due to the Coanda effect, which is illustrated with a spoon experiment.

In a classroom-type of experiment, water falling from a tap is passively re-directed by the convex curved side of a spoon due to the Coanda effect (towards the right). The spoon is observed to pivot sideways in the opposite direction (towards the left), and to remain suspended in the water-flow.

According to Newtonian mechanics, the change in direction of the waterflow causes a change in its momentum, which creates a turning force. The reactive, equal and opposite force pushes the spoon sideways to the left and slightly downwards. As long as the spoon re-directs the water-flow, then it remains suspended in the water. See Fig. 2c-i.

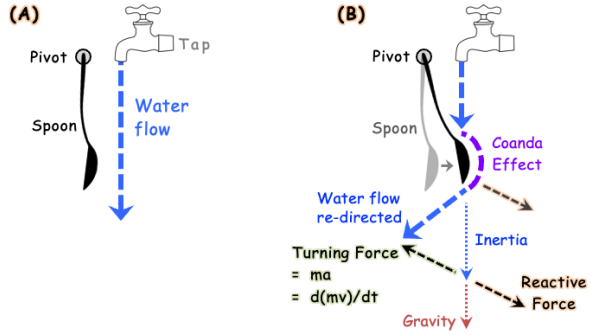


Fig. 2c-i. Spoon experiment demonstrating the Coanda effect.

The turning force generated in the spoon experiment is not significant to the analysis of the lift actively generated by a wing. But it is significant as a minor force passively generated by a boat sailing into the wind.

The change in momentum can be calculated using the standard Newtonian equation based on the mass (m) and velocity (v) of the water re-directed each second (dt); as summarized by the equation: Force = ma = d(mv)/dt.

The more that the water-flow is re-directed by the spoon (to the right), due to increased water-flow speed. Then the greater the turning and reaction forces generated causing the spoon to pivot in the opposite direction (to the left).

The spoon is not affected by Bernoulli's principles of fluid dynamics or the Venturi effect. When the water-flow is re-directed, there is no noticeable change in water-flow velocity.

The spoon experiment in more detail:

1) Experimental conditions: Laminar water-flow is directed downwards from a water tap. A common tablespoon hangs freely from a pivot close to the water-flow, just 5 millimeters apart. The spoon is able to swing from side-to-side while attached to the pivot.

2) When hanging freely by the side of the water-flow in (A), the spoon is unaffected by the speed or volume of the water-flow.

3) When the curved side of the spoon is pushed gently into the descending water-flow, while attached to the pivot. Then the water-flow is re-directed (to the right) by the spoon to the right due to the Coanda effect. This action causes the spoon to be pulled diagonally downward and sideways in the opposite direction (to the left). The spoon remains suspended in the water-flow.

4) Only if the water-flow is turned off does the spoon swing back to its original position, due to gravity. These observations are only achieved if the water-flow is sufficiently fast and voluminous, and the spoon is sufficiently large with enough curvature. Otherwise, the spoon does not remain attached to the water-flow.

5) If the spoon's curvature is reversed, or if other non-curved objects such as a knife are placed in the water-flow, while connected to a pivot. Then the water-flow simply pushes these objects out of the water-flow (to the right). The objects return immediately to their original position, out of the water's path downwards. The objects do not remain suspended in the water-flow like the spoon does, as described above.

6) A common explanation of this experiment is that the spoon remains attached to the water-flow due to the water's surface tension; water is 'sticky'. However, this explanation is not an accurate depiction of what is observed. Water's 'stickiness' may explain the Coanda effect, but not the resultant forces arising.

7) Increasing the volume and speed of the downward water-flow increases the Coanda effect and the reactive force.

Similar experiments – Ping-Pong ball

Similar experiments with curved objects, such as balls and wings, have produced comparable results. However, researchers provided different non-Newtonian explanations for the forces involved. [18][19] See Fig. 2c-ii.

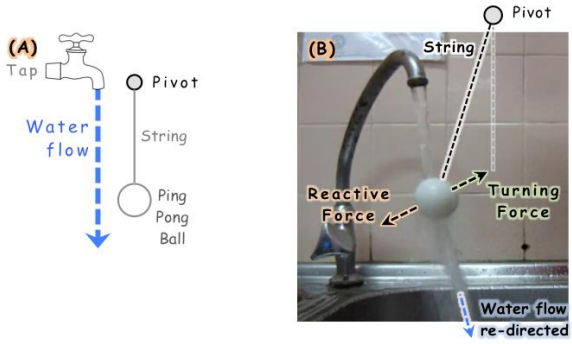


Fig. 2c-ii. Coanda effect – ping pong ball experiment. [19]

In the Ping-Pong ball experiment above, it is worth noting that the string attached to a pivot and Ping-Pong ball remain taut when the ball is suspended in the water-flow. This dynamic indicates that the reactive force is pulling the ball diagonally downwards to the left, towards the ground. In other words, the reactive force is **not** pulling the ball upwards. The ball only swings slightly upwards relative to its starting position due to the mechanics of the pivot and string. The rigid string prevents the ball from moving downwards.

This aspect can be tested by using a rubber band attached to the ball, instead of a string. In this case, the rubber band allows the water-flow to pull the ball to the left and diagonally downwards towards the ground.

D. Evidence for the Coanda effect on wings.

Wind tunnel experiments

Wind tunnel experiments demonstrate relative airflows over an airplane wing. The upper airflow above the wing is re-directed downwards along the topside of a curved wing, due to the Coanda effect. In the absence of a sufficiently strong Coanda effect, airflow separation can arise, which causes turbulence. The wing's flat underside simply deflects the air downwards. See Fig. 2d-i.

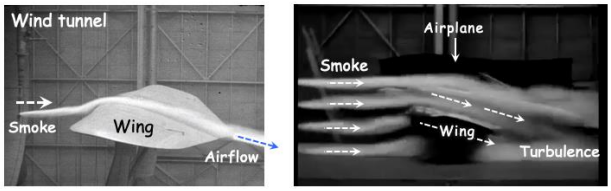


Fig. 2d-i. Relative airflows passing over wings. [10]

The amount and velocity of the airflow re-directed by the topside of the wing due to the Coanda effect, depends on the maintenance of laminar (non-turbulent) airflow attached to the wing. Airflow separation causes turbulence and reduces the airflow re-directed by the wing. See Fig. 2d-ii.

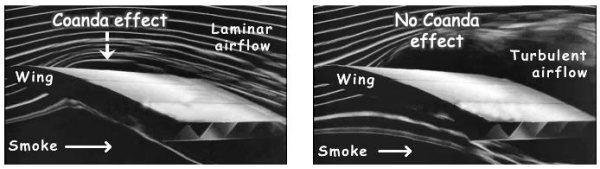


Fig. 2d-ii. Smooth vs. turbulent airflows on a wing. [10]

In general, wings produce a stronger Coanda effect with laminar (non-turbulent) airflow at a lower wing AOA, higher airspeed, and where the wings are deepest (largest chord, such as near the fuselage). Conversely, the Coanda effect is weakest at high AOA, slower airspeeds, and where the wings are narrow (i.e. short chord).

Wing vortices show the Coanda effect

Vortices formed on airliner engines show how the static air is displaced as a moving wing passes through it, helped by the Coanda effect. See Fig. 2d-iii.



Fig. 2d-iii. Static air accelerated over a moving wing. [20]

Coanda effect and wings

Some fighter jet wing and fuselage designs show pronounced curvature that maximizes the Coanda effect. The Coanda effect helps to explain why airplanes keep the topside of wings clear of any obstructions that could disrupt airflows. See Fig. 2d-iv.



Fig. 2d-iv. Fighter jets. [13] [20]

Similar to the spoon and water experiment above, a simple classroom-type of experiment demonstrates how the Coanda effect on the topside of the wing. The wing re-directs a relative airflow from a hair dryer, causing the wing to rise upwards.

The paper wing has a pronounced curved topside and a flat underside, with a vertical stabilizing tail. There is no fuselage or tail section present. The wing is held in place by a vertical string at the leading-edge. The string allows the wing to move vertically up and down, but prevents it moving horizontally forwards or backwards. [15] See Fig. 2d-v.

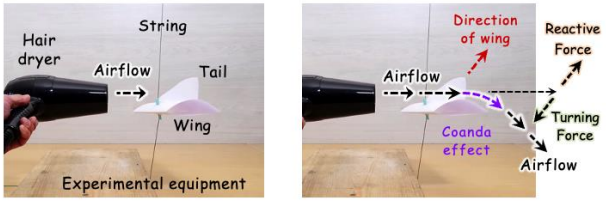


Fig. 2d-v. Wing and hair dryer experiment. [15]

Many people mistake this experiment to demonstrate how a wing generates lift due to low air pressure on the topside of the wing, caused by a relative airflow. This is false. [6]

Rounded vs. Sharp edges

The Coanda effect on the topside of the wing, due to wing curvature is extremely significant. For example, the Coanda effect explains why an evenly curved (rounded) surface produces significant downwash and lift. In contrast, a pointed (sharp) surface produces negative lift. This difference applies to both a stationary wing exposed to relative airflows, as well as a wing moving through static air. See Fig. 2d-vi.

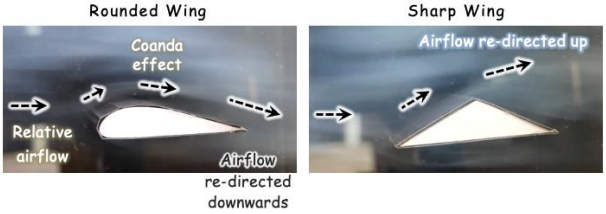


Fig. 2d-vi. Lift generation by rounded and sharp surfaces exposed to a relative airflow compared.

E. The Bernoulli paper experiment. [6]

The Bernoulli paper experiment is another simple classroom-type experiment in which a person blows across the top of a flat piece of paper, which rises upwards to be horizontally aligned. The paper is initially hanging downwards (vertically aligned). See Fig. 2e-i.

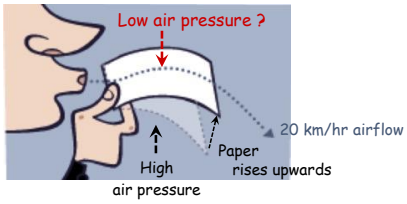


Fig. 2e-i. Bernoulli paper experiment.

The logic presented above for the wind tunnel, wing and ball, as well as the hair dryer and scale experiments can also help explain the Bernoulli paper experiment. Namely, the high-velocity airflow above the paper creates low air pressure. The high pressure beneath the paper then **pushes** the paper upwards, towards the low-pressure airflow.

This explanation is very different to the common explanation, where the paper is **pulled** upwards by the low-pressure airflow, according to Bernoulli's principles of fluid dynamics.

Problems using Bernoulli to explain lift

This experiment has been incorrectly used to demonstrate how a wing generates lift for many years, as pointed out by academics as far back as 1972. [21]

Irrespective of whether the paper is pushed or pulled upwards, some reasons why the Bernoulli paper experiment cannot explain the lift generated by a wing include:

- The relative airflow only pulls the rear of the paper (or trailing edge of a wing) upwards, to the level of the front part of the paper (or the leading edge of the wing). There is no process for the front part of the paper to rise up.

  This means that according to the Bernoulli paper experiment, the wing would tilt to a flat AOA, and the front of the wing would never rise upwards. An aircraft would never leave the runway.

- Bernoulli's principles and equations of fluid dynamics only apply to enclosed environments, such as water pipes used for plumbing. These principles and equations do not apply to an open atmosphere for a wing in flight.

- A back-of-the-envelope type of calculation of the lift generated, described below, shows that very little lift force is generated by blowing across a piece of paper. It is not feasible for a relative airflow difference above and below the wings to produce the amount of low pressure required to produce sufficient lift for an airplane to fly.

Alternative explanation – A change in momentum

The airflow re-directed downwards over the top of the paper, can be explained by the Coanda effect; as fluid flow naturally follows a curved surface. Consequently, the Bernoulli paper experiment can be explained by a change in momentum of the airflow, as the airflow is re-directed down.

This Newtonian approach is similar to the water and spoon experiment: Water falling from a tap is passively re-directed to the right (and slightly upwards) by the curved side of a spoon due to the Coanda effect.

According to Newtonian mechanics, this action creates a small turning force, due to the change in momentum of the water flow. The reactive equal and opposite force pushes the spoon sideways to the left (and slightly downwards). The same logic can explain the paper rising upwards in the Bernoulli paper experiment. See Fig. 2e-(i-iv).

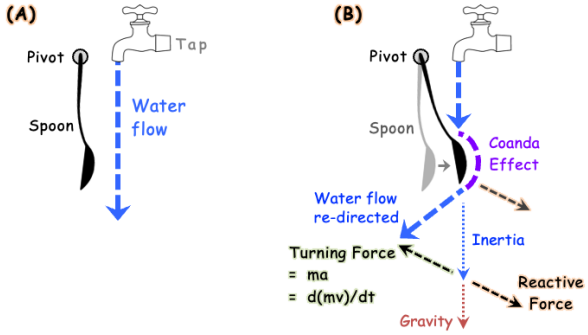


Fig. 2e-ii. Spoon experiment demonstrating the Coanda effect and a change in momentum of the water flow.

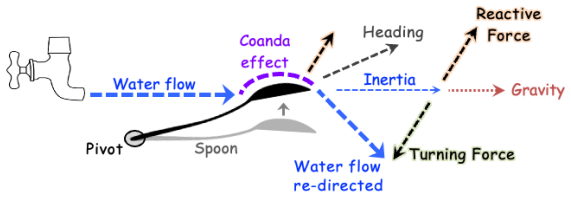


Fig. 2e-iii. Spoon experiment demonstrating the Coanda effect and a change in momentum of the water flow
– rotated 90° to the horizontal direction.

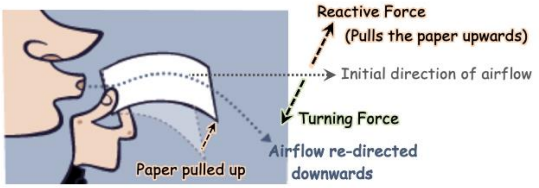


Fig. 2e-iv. Bernoulli paper experiment attributed to a change in momentum of the airflow.

3. SWIMMING OUT

A. *The physics of swimming out.*

Where:
- m = Mass of water passed through and pushed back.
- m/dt = Mass flow rate.
- dt = Change in time (per second).
- dv = Change in velocity (v) of the water.
- v = Velocity of the water displaced back by the fish.
- a = dv/dt = Acceleration.

Equations:
- Force = ma = m * dv/dt = m/dt * dv [1]
- Force = ma = m * dv/dt = d(mv)/dt [1]
- Momentum = mv [1]

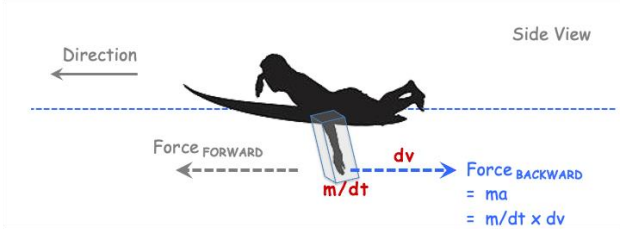


Fig. 3a. Forces acting on a person's feet swimming.

The physics of swimming is straightforward. The force (Force = ma) generated by a swimmer depends on the mass of water accelerated backwards by the feet and legs. Except, a surfer typically does not kick with their feet. See Fig. 3a.

More precisely, a surfer's arms accelerate a mass of water each second (m/dt) backwards to a velocity (dv), which creates a backward force as described by the equation:

$$Force_{BACK} = ma = m/dt * dv \quad (1)$$

The inertia of the air provides resistance to the downward force. This dynamic allows for the generation of a reactive equal and opposite upward force:

$$Force_{BACK} = Force_{FORWARD} \quad (2)$$

Equations (1) and (2) can be combined as follows:

$$Force_{BACK} = Force_{FORWARD} = m/dt * dv \quad (3)$$

$$Or\ simply: \quad Force_{FORWARD} = m/dt * dv \quad (4)$$

$$Units: \quad N = kg/s * m/s$$

The swimmer's efficiency at generating the optimal forward force depends on finding a balance between 'm/dt' and 'dv', and directing the fluid flow directly backwards (not sideways or downwards).

See the report: How fish swim according to Newtonian physics, and how to win a swimming race. [2] In short, the best techniques to swim faster include:

1) Minimizing drag. The surfboard minimizes the drag from the surfer's body, as compared to a swimmer.
2) Maximizing the Coanda effect.
3) Optimizing the angle-of-attack (AOA).
4) Maximize 'm/dt' and 'dv', and therefore, the force generated (Force = m/dt * dv). For example, flippers help because they increase 'm/dt' with each kick.

In these equations above the increased velocity of the water is expressed as 'dv', and not as acceleration ('dv/dt'), because this action is not time-dependent. It is due to a one-off force (impulse) from the swimmer. In contrast, the mass of water passed through by the swimmer is time-dependent, and therefore, is expressed as the mass flow rate (m/dt).

The Newtonian explanation is not the prevailing method used by biologists to explain the physics of swimming, which includes the Elongated Body Theory and The Resistive Force Theory for undulatory locomotion. [3]

B. *Transfer of momentum.*

There is no net gain or loss of momentum, energy and mass in this process of generating a forward force. Momentum and kinetic energy is transferred from the swimmer to the water, by accelerating the water backwards. See Fig. 3b

$$Force_{BACK} = ma = d(mv/dt) \quad (5)$$

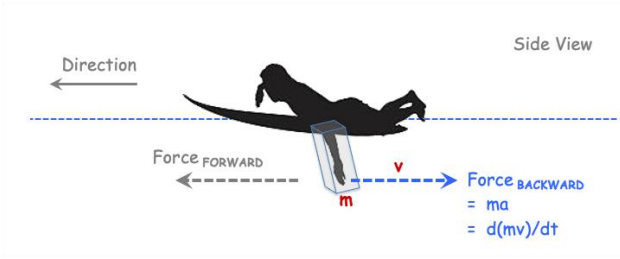


Fig. 3b. Momentum theory of swimming.

Combining equations (2) and (5) allows the forward force to be expressed as the change in momentum of the water:

$$Force_{BACK} = Force_{FORWARD} = d(mv)/dt \quad (6)$$

$$Or\ simply: \quad Force_{FORWARD} = d(mv)/dt \quad (7)$$

$$Units: \quad N = (kg * m/s)/s$$

C. *Two Newtonian equations for swimming.*

The analysis above provides two Newtonian methods and equations to calculate the forward force generated by a wing:

$$Force_{FORWARD} = ma = m/dt * dv \quad (mass\ flow\ rate) \quad (4)$$

$$Force_{FORWARD} = ma = d(mv)/dt \quad (momentum\ theory) \quad (7)$$

Both equations (4) and (7) are based on Newtons 2nd Law of Motion (Force = ma). Both are correct and produce the same values, but express the same thing slightly differently.

Comment: This is NOT the prevailing view that explains the forces generated by swimming. The physics of how people and fish swim remains an enigma.

4. RIDING A WAVE

A. *The physics of riding a wave.*

Surfers riding a wave generate speed by taking advantage of the water ascending from the trough of the wave. See Fig. 4a-i.

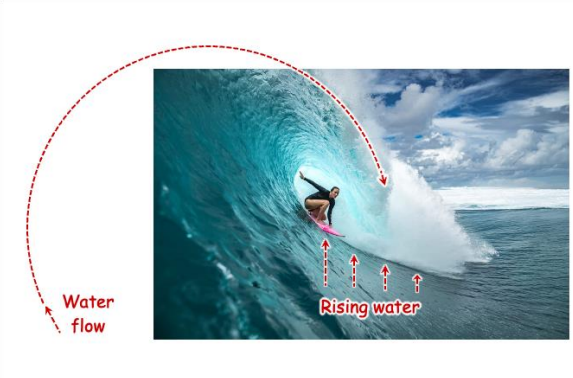


Fig. 4a-i. Water flow on a breaking wave.

Two enigmas

Newtonian mechanics can be applied to explain two surfing enigmas that arise when riding a wave: See Fig. 4a-ii.

1) The surfer can travel laterally across the wave at speeds several times faster than the speed of the wave.

2) The water appears to grip the edge of the surfboard, pulling it downwards and enabling 'nose-riding', as described below in Section 5.A on page 11.



Fig. 4a-ii. Surfing enigmas.

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B. *Change in momentum of the water in a wave.*

The movement of the wave can be described based on the change in momentum of the water, based on the standard Newtonian equation for a change in momentum:

$$\begin{aligned} \text{Force} &= ma \\ &= m * dv/dt \\ &= d(mv)/dt \end{aligned}$$

The momentum of the wave depends on the mass (m) and velocity (v) of the water. The greater the mass or velocity of the water, then the greater the change in momentum.

The momentum of a wave also depends on the frequency and amplitude of the wave. Increased amplitude or frequency of the wave increases the wave's momentum. See Fig. 4b-i.

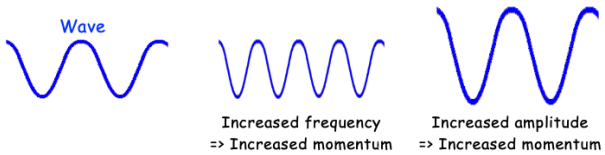


Fig. 4b-i. Forces acting on a falling bead chain.

An analysis of the water's momentum: See Fig. 4b-i.

- As the water starts to rise upwards from the trough of the wave it has low vertical velocity, and therefore, it has little vertical momentum.
- At the mid-point of the wave, the water has the maximum vertical velocity, and hence, it has maximum vertical momentum.
- At the crest of the wave, the water again has low velocity, and therefore, it has little momentum.

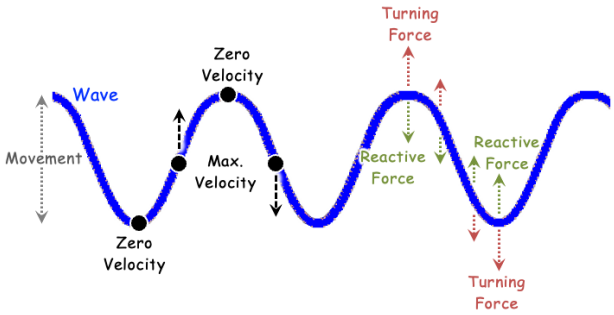


Fig. 4b-i. Forces acting on a falling bead chain.

Therefore, the ideal position for a surfer to ride a wave is at the mid-point, where the wave's upward velocity is greatest. See Fig. 4c-iv below.

At the trough and the crest of the wave, the change in direction of the water causes a change in momentum (turning force), due to inertia. The reactive, equal, and opposite force generated acts on the water, causing it change direction.

C. Surfing across a wave (riding a wave).

In short, the rising water of the wave pushes the surfboard upward. More precisely, the rising mass of water each second (m/dt) of the wave decelerates (dv) on contact with the surfboard. This action generates an upward force (Force $_{UP}$) that pushes the surfer upward, as shown by the equation: See Fig. 4c-(i-ii).

Force $_{UP}$ = m/dt * dv

Momentum is transferred from the wave to the surfer, by the surfboard slowing down the rising water. Gravity constantly pulls the surfer downward.

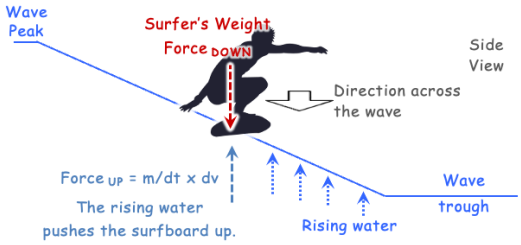


Fig. 4c-i. Forces acting on a surfboard riding a wave. 1

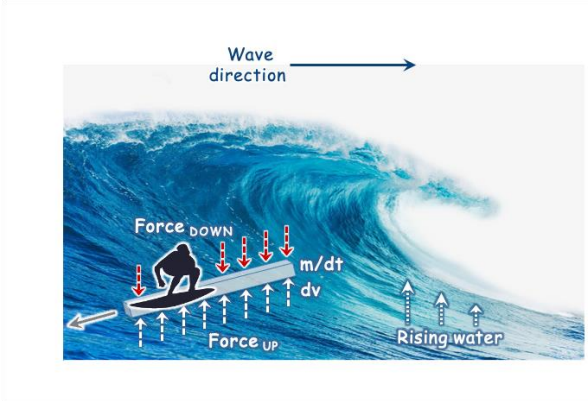


Fig. 4c-ii. Forces acting on a surfboard riding a wave. 2

m/dt (mass flow rate)

The mass of water each second (m/dt) in contact with the surfboard depends on factors such as:

- The surface area of the surfboard that is in contact with the wave. Larger surfboards can generate larger forces.

  However, the size of the surfboard does not affect the deceleration of the rising water.
- The speed of the wave. A faster wave pushes a greater mass of water against the surfboard each second.
- The speed of the board. A faster surfboard travels through a greater mass of water in the wave each second.
- The AOA of the surfboard against the wave.
- The mass of water displaced by a surfboard can be seen by the trench in the wake behind the surfboard . See Fig. 4a-ii.

dv (change in velocity)

The velocity of the water decelerated (dv) by the surfboard depends on factors such as:

- The larger (taller) and faster moving the wave. Then the greater the deceleration (dv) of the rising water and therefore, the greater the force that can be generated by a surfboard.
- The water rises fastest near the middle section of the wave, and therefore, can generate the greatest force.
- Above the force generated is described by the surfboard decelerating the rising water. However, the surfboard may accelerate the rising water in the opposite direction (i.e. downward), which would generate a greater upward force.
- The speed and mass (i.e. momentum) of the surfer.

  A faster-moving surfboard with a heavier surfer has greater momentum. In turn, this means that the surfboard accelerates the water away from the wave at a higher velocity. The surfboard is hitting each water particle with greater momentum and force.
- The AOA of the surfboard against the wave.

'm/dt' and 'dv' are related but separate factors. The forces generated depends on how efficiently these factors are combined. The surfer controls the angle and direction that the forces are generated to provide stability and maneuverability.

The upward force

The upward force generated by the surfboard is applied by the surfer to:

- Support the surfer's weight and prevent the surfboard from sinking into the water.
- To constantly move forward with the wave, maintaining the surfer's position on the rising part of the wave.

  This explains why surfboards are typically pointed forward on small and slowing moving waves. In contrast, on large and fast moving waves can be used to generate a larger force. Hence, the surfboard can be pointed more laterally across the wave See Fig. 4c-iii.
- To move up or down the wave, as well as to move laterally along the wave.

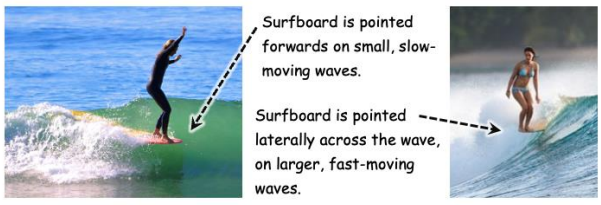


Fig. 4c-iii. Surfboard direction compared for different wave sizes.

Additional considerations

Additional considerations include:

- A surfer riding a wave is merely maintaining their speed towards the beach relative to the wave.
- A surfer riding a wave and taking a direct line to the beach is not moving relative to the wave.
- A surfer cannot arrive at the beach much before the wave.
- A surfer riding a wave, as seen from the beach appears to traverse laterally across the wave with the surfboard pointed down.

  As the wave constantly pushes the surfer upwards, they approximately maintain their altitude (relative position and height on the wave), despite gravity pulling the surfer downward. See Fig. 4c-iv.

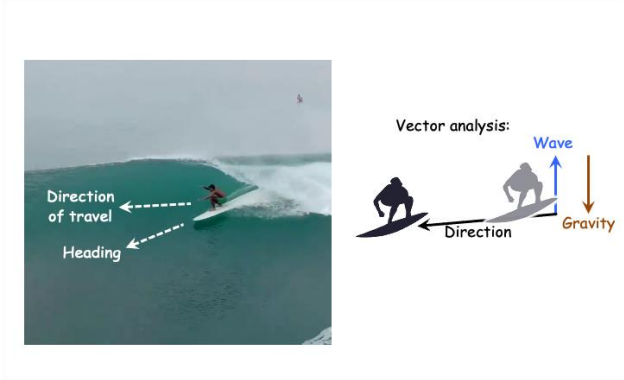


Fig. 4c-iv. A surfer riding a wave, as seen from the beach.

- The greater the downward and upward forces generated, then the faster the surfboard can travel along the wave, while maintaining its relative position on the wave. However, if the surfer travels too fast, they could outrun the wave.
- There are various trade-offs between 'm/dt' and 'dv'.

  For example, a light surfer (low mass) can only exert a small downward force on the rising water of the wave due to gravity and their low weight. As a result, light surfers generate slower speeds surfing laterally.

D. Angle-of-attack (AOA).

The conventional approach to explain AOA by the angle between the (incoming) relative water flow and the surfboard. See Fig. 4d-(i-ii).

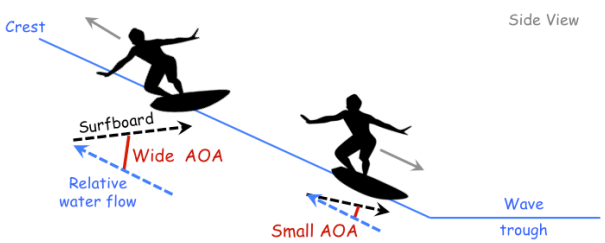


Fig. 4d-i. Conventional AOA of a surfboard.



Fig. 4d-ii. AOA of a surfboard.

By adjusting the surface area of the surfboard in contact with the rising water of the wave, and adjusting the surfboard's AOA. The surfer can control the amount and the directions of force generated.

More precisely, altering the AOA allows the surfer to alter the direction that the rising water is decelerated (or accelerated). This action then alters the direction of the upward force generated.

For example, a wide AOA means that the surfboard pushes the water in the direction more towards the wave trough. In turn, the equal and opposite upward force then is in the direction of the crest, and the surfboard ascends the wave. See Fig. 4d-iii.

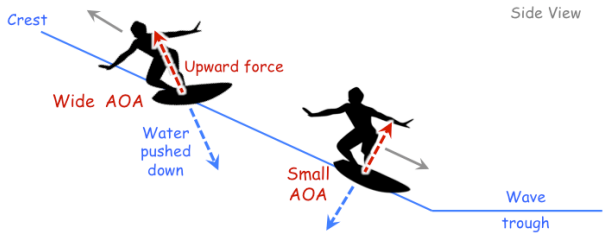


Fig. 4d-iii. Upward forces and AOA.

For example, beginner surfers use large surfboards. This can be explained by the lack of skill at controlling the AOA and the direction that the rising water is pushed.

Altering the AOA can be used to adjust the surfer's position on the wave, and to accelerate laterally across the wave.

In addition, different AOAs alter the efficiency that forces are generated, by altering the mix of 'm/dt' and 'dv' produced. This is partly because other variables also change; such as the surface area of the surfboard in contact with the water.

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E. Wave riding behind a boat.

Riding the waves directly behind motorboats demonstrates the physics of surfing. The propellers at the rear of the motorboat accelerate the static water backwards, creating a wave. The surfer can ride the wave similar to a 'normal' wave on a beach. Specifically, the force from the ascending water behind the motorboat pushes the surfer upwards. The surfer uses gravity to pull them down and forwards. See Fig. 4e.



Fig. 4e. Riding a wave behind a motorboat. [17]

The surfer steals momentum from the wave by slowing the water down, which is converted into lift and forward motion. The surfer alters their AOA to select the desired mix of forward motion and lift up and down the wave.

F. Surfing backwards on a river.

The same Newtonian logic that explains surfing on an ocean wave, can be applied to explain the physics of surfing upstream on a river, against the waterflow.

For example, people can be seen surfing on a stationary wave in a river, as the river descends towards the ocean. The surfer can maintain their position indefinitely on the upstream side of the river wave. A stable position is achieved by selecting an appropriate AOA on the rising waterflow of the wave. The force of the river (Force $_{RIVER}$) pushing the surfer upward and backward, equals the downward force exerted by the surfer's mass due to gravity (weight). See Fig. 4f.



Fig. 4f. Riding a wave on a river against the waterflow.

G. Additional comments.

To catch a wave a surfer needs to have a similar speed to the wave. The ascending water at the front of a breaking wave helps by accelerating the surfer forwards from the wave trough. If the surfer hasn't accelerated to the wave's speed by the time the crest reaches them, then the surfer fails to catch the wave.

The wave is pushing the surfboard forwards with it as it travels towards the beach. This means that:

- The greater the surface area of the surfboard in contact with the wave, the greater the force exerted by the wave against the board (Force = Pressure * Area).
- For a breaking wave, the closer the surfboard is to the top of the wave, then the stronger the force on the board from the water circulating forwards.
- Conversely, the closer to the trough of the wave, then the least exposure to the water circulating forwards.
- For a breaking wave, the steepest point of the wave is where the water is rising the fastest. Therefore, this point provides the greatest force from the wave on the surfboard.
- If the surfer riding a wave descends or ascends the wave, then they gain/lose speed and kinetic/potential energy due to gravity
- For a surfer riding a wave, momentum is transferred from the wave to the surfer.
- Drag is a key factor that restrains a surfer's speed from increasing too much.

H. Increased buoyancy.

The upward force gained by the surfboard from riding a wave means that the surfer and surfboard are pushed up towards the surface of the water.

This dynamic is particularly noticeable on small short boards, which provide little buoyancy. A surfer and surfboard may be partially submerged while waiting for a wave. But the surfer is able to stand on the board and skim across the surface of the water once they gain enough speed. See Fig. 4h.



Fig. 4h. Partially submerged surfers waiting for a wave.

5. The Gripping Force

A. *Nose-riding.*

The surfboard is known to grip the water to permit nose-riding and surfing on one edge in a barrel. The surfboard appears to be suspended on the water with the surfer in an unbalanced position. See Fig. 5a-(i-ii).



Fig. 5a-i. Nose-riding.



Fig. 5a-ii. Newtonian forces acting on a surfboard nose riding and surfing in a barrel.

At first glance, nose-riding and the surfer's position surfing a barrel appears illogical, as the surfer is standing in an unbalanced position on the surfboard, as compared to its center of gravity when not riding a wave. It appears that the surfboard should be unbalanced and tip over, but it doesn't.

This paper asserts that the board grips the water primarily due to curvature on the edge of the surfboard that causes the Coanda effect, as described above. In addition, fin alignment can boost the gripping force. These create forces that pull the surfboard towards the wave.

Nose riding and these other positions require a minimum speed. It typically cannot be done on a shallow wave or at a low speed. It cannot be done if the surfboard is stationary.

The force produced by the Coanda effect to grip the surfboard to the water depends on 'm/dt' and 'dv'. In turn, this depends on the speed, length, and curvature of the edges of the surfboard.

Long boards are favored for nose-riding because they provide longer edges to grip the water, and therefore, generate a greater force.

Nose-riding typically involves both edges of the surfboard in contact with the water. This generates a stronger Coanda effect, as compared to only one edge being in contact with the wave.

B. *The Coanda effect.*

Fluid flow naturally follows a curved surface due to the Coanda effect. For example, water falling from a tap is re-directed by the curved side of a spoon demonstrating the Coanda effect, as illustrated in Fig. 5b-i.

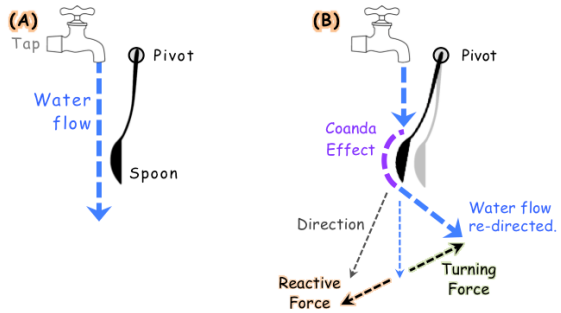


Fig. 5b-i. Coanda effect – Spoon experiment.

According to Newtonian mechanics, the water flow passively re-directed by a spoon due to the Coanda effect creating a small turning force due to the change in momentum of the water flow. The reactive equal and opposite force pushes the spoon diagonally to the left sideways and downward. However, the spoon pivots to the left as far as the reactive force allows.

The change in momentum can be calculated using the standard Newtonian equation based on the mass (m) and velocity (v) of the water re-directed each second (dt); as summarized by the equation:

$$\text{Force} = ma = d(mv)/dt$$

For the surfboard riding a wave, the direction of the water flow is upward, which is the reverse of the spoon experiment above where the water flow is downward. However, the physics is the same. See Fig. 5b-ii.

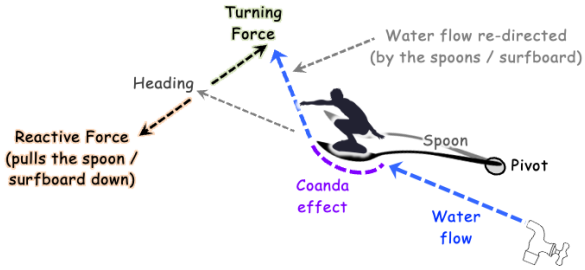


Fig. 5b-ii. Gripping force arising from the Coanda effect.

The greater the curvature on the edge of a surfboard, then the stronger the Coanda effect. Then the better that the surfboard grips the water.

The gripping force is described by the same Newtonian equation (Force = m/dt * dv).

The 'm/dt' and 'dv' components of the gripping force differ to those of the downward force above. The mass and velocity of the water re-directed by the Coanda effect are different to the mass and velocity of the water deflected by the surfboard.

Similar to the downward force, the faster the surfboard is going, then the greater 'm/dt' and 'dv', and the stronger the Coanda effect.

The Coanda effect explains nose-riding ad how surfers can stand in seemingly unbalanced positions when riding a wave.

C. Fin alignment.

In brief, if the fins on the rear underside of the surfboard are aligned to push water up and outwards, then they boost the gripping force created by water flow. See Fig. 5c.

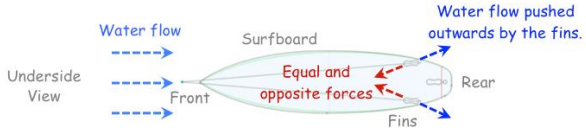


Fig. 5c. Example of a surfboard fin alignment.

This paper considers fins to be secondary importance to the Coanda effect also because:

- Fins' primary role is to improve the surfboards stability and maneuverability; and not as a gripping force
- Adding fins to a surfboard improves traction and the gripping force, but not all that much. Whereas the gripping force is known to be strongly correlated to the curvature of surfboard edges.
- Fins tend to be aligned vertically. This means that the fins do not push water upwards, to create an equal and opposite downward force on the surfboard.

D. Nose-riding – other explanations.

How the water grips the board to enable nose-riding is typically explained by:

- The surface tension of the water with the surfboard.
- The weight of the water on the back of the board.
- Fluid mechanics (i.e. hydrodynamics and Bernoulli).
- Buoyancy of the surfboard.

None of these explanations are satisfactory or accurate as a complete explanation. In addition, these have not been proved by experimentation or mathematical calculation as the determining factor of how a surfboard grips the water.

At best they are only partially correct and do contribute to the solution. These are not the key-determining factor that explains how a boat grips the water.

It is beyond the scope of this paper to discuss these other explanations in any detail.

There appears to be agreement among surfers that nose-riding is possible because a force is pulling the rear of the surfboard downwards. But there is no agreement on the physics involved.

E. Leverage, pivots, and moments – the basics.

For reference, the basics of leverage, pivots, moments are reviewed below for the avoidance of confusion.

Leverage can create torque (or moment) around a pivot which amplifies or dilutes a force, depending on the perspective. Force is measured in 'N' while leverage is measured in 'Nm'. Torque (or the moment) is calculated using the distance from the pivot multiplied by the force, as described by the equation: See Fig 5e.

Moment (Nm) = Force (N) * distance (m)

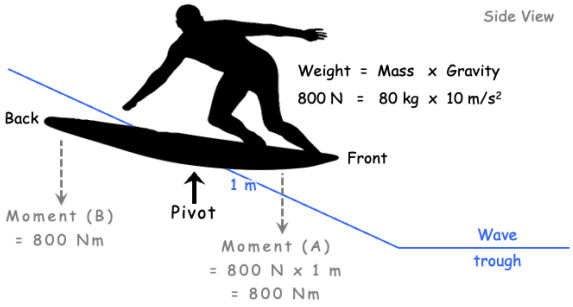


Fig. 5e. Moments and forces acting on a pivot.

For example, the leverage of a 80 kg surfer standing on the nose of a surfboard, 1 meter from the surfboard's pivot point on the water, which produces a moment of about 800 Nm. This calculation assumes gravity to be about 10 m/s^2. This is a relatively large force for a surfboard. How the surfer positions their weight is critical to the surfboard's balance.

For the surfboard to maintain its balance on the wave, the 800 Nm moment (A) due to the surfer nose-riding (calculated above), requires a compensating 800 Nm moment (B) on the opposite side of the pivot at the back of the surfboard. This compensating 800 Nm moment (B) comes from the Coanda effect gripping the surfboard.

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6. Discussion of Results

A. Comment.

In contrast to other theories, Newtonian physics provides a simple and easy-to-understand explanation of surfing that fits with what is observed in practice.

This analysis explains long-standing puzzles that have eluded solution. It provides new insights that are useful to the niche group of surfers and physicists. It is significant as it allows a better understanding the dynamics of surfing, and therefore, to design better surfboards and to become better surfers. This knowledge may have commercial applications outside of surfing.

It is puzzling that no one has attempted to use Newtonian physics to explain surfing previously. It is also puzzling that few surfers appear concerned that this is worth solving, being content to learn and progress by trial-and-error.

The analysis indicates that by introducing backward-orientated, diagonal groves in the surfboard's edge (rail) may improve its forward speed. This is because the Coanda effect could push more water flowing along the edges backwards, rather than just to the side of the surfboard as currently occurs.

B. Slope soaring by gliders.

The principles of slope soaring by gliders along hilltops are also similar to wave surfing. In slope soaring, the rising mass of air in the wind pushes the glider up and backwards.. See Fig. 4h-i.

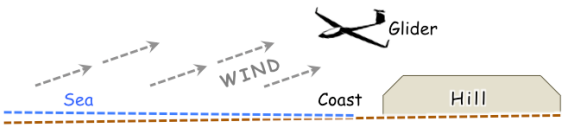


Fig. 6a. Slope soaring along a coastline.

The glider steals momentum from the wind, which it converts into lift and forward motion. This allows the glider to gain altitude and airspeed.

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7. Conclusions

Newtonian mechanics provides a straightforward explanation of the physics of surfing, consistent with the forces generated by the observed fluid flows. In particular, the Newtonian approach solves the enigmas:

- How a surfer can surf at speeds several times faster than the wave that they are riding; and
- How the water grips the surfboard due to the Coanda effect.

8. Additional Information

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Author Contributions: This paper is entirely the work of the author, Nicholas Landell-Mills.

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Acknowledgements: None.

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- Or buy me a coffee: https://bmc.link/zhJIg4zRCW

Thank you in advance for any contributions, which will encourage further research!

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